☕ The Great Coffee Race

Which Cup Keeps Your Drink Warm the Longest?

A step-by-step Python experiment explained so simply that **anyone** can understand how thermal insulation, lids, and evaporation work!

Real Physics Data Easy Python Code 4 Visual Graphs

Swipe to Start ➡️

MEET THE COMPETITORS

3 Cups Enter the 50-Minute Cooling Race



Cup A: Insulated (No Lid)

"The Winter Coat with No Hat"

Has thick steel walls to stop heat from escaping the sides, but heat can still fly straight out the open top!



Cup B: Insulated (With Lid)

"Cozy Coat + Winter Hat"

Has thick double walls AND a tight lid on top. Traps both warmth and steam so nothing escapes easily.



Cup C: Ceramic Mug

"The Thin T-Shirt"

A classic mug. Clay walls absorb heat quickly and pass it out into the cool room air from every direction.

⏱️ **The Rule:** We measured the temperature of all three cups every 5 minutes for a total of 50 minutes.

How Fast is Heat Escaping? (`.diff()`)

Instead of just asking *"How hot is the coffee right now?"*, we ask our computer helper (Pandas) to tell us **how many degrees were lost during each 5-minute jump** using `.diff()`

```
# Ask Python for step-by-step heat loss
 temp_drops = df[['insulated no lid', 'insulated',
'ceramic']].diff()
```

💨 The Great Escape (Steam Power)

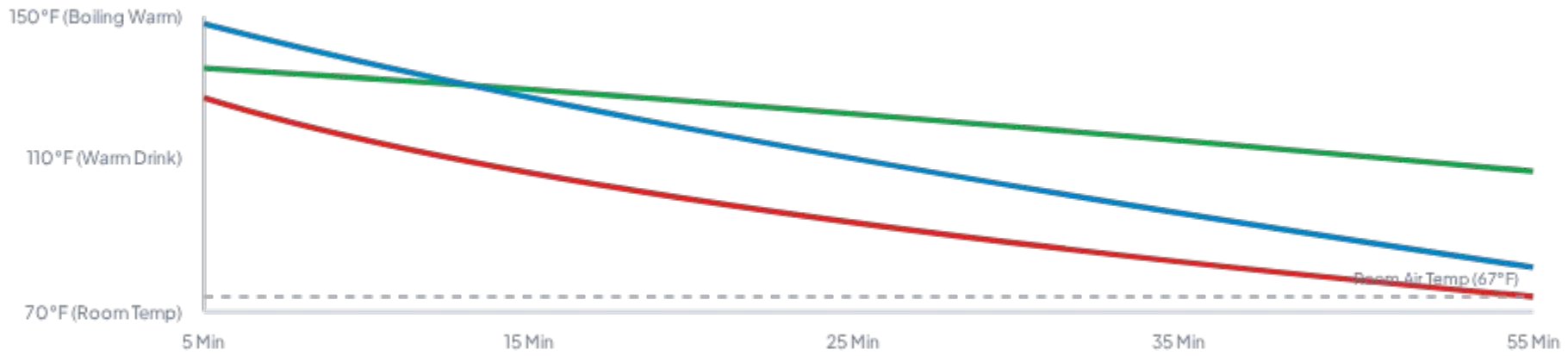
In the first 5 minutes, open cups lose up to **8.4°F** instantly because steam flies straight into the air!

🧢 The Lid Barrier

The lid blocks steam, cutting initial heat loss by **over 60%**. Steam bounces right back into the liquid!

💡 **Simple Analogy:** Imagine running down a hill. `.diff()` measures how many feet you drop with every single step you take!

Line Chart: Watching the Coffee Slide Downhill



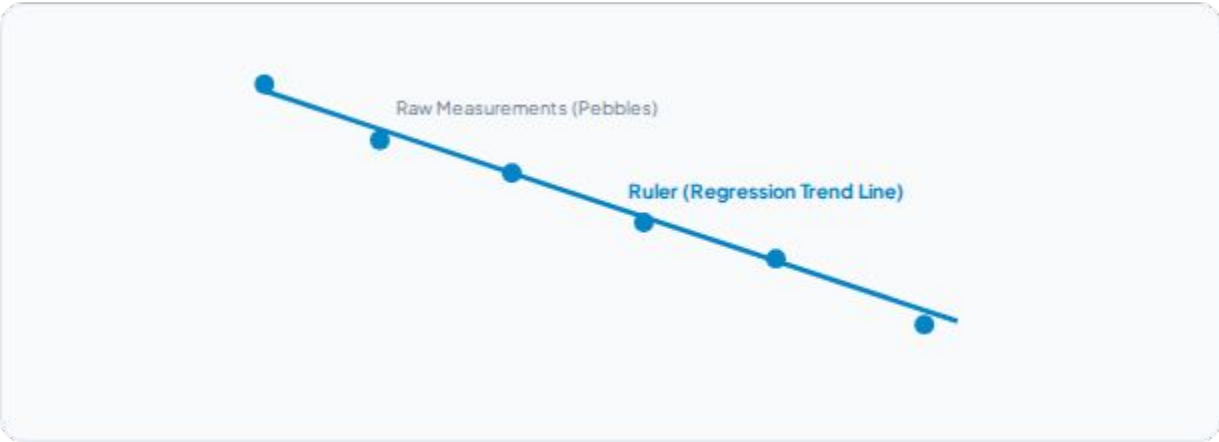
🟩 Lidded Cup: Gentle Gentle Slope 🟢

🟦 No Lid: Medium Drop 🔵

🟥 Ceramic: Fast Steep Slide 🔴

STEP 3 • FINDING THE TREND

Scatter Plot: Pebbles & the Straight Ruler



1. The Pebbles (Scatter Dots)

These are the exact thermometer readings taken at each minute. Real life is slightly bumpy and imperfect!

2. The Ruler (Regression Line)

A smooth line drawn straight through the middle of the pebbles to show the main overall direction.

🎯 **Why do both?** The dots show real facts; the line shows us the general pattern!

STEP 4 • THERMAL COLORS

Heatmap: The Weather Map of Coffee

Time	Insulated No Lid	Insulated Lid	Ceramic
5 Min	146.6°F 🔴	135.5°F 🔴	128.8°F 🔴
25 Min	109.8°F 🟠	121.1°F 🟠	96.1°F 🟠
55 Min	83.7°F 🔵	103.8°F 🔵	77.0°F 🔵

🔴 Hot Red Boxes (Top)

Shows where the coffee starts blazing hot right out of the pot!

🔵 Cool Blue Boxes (Bottom)

Shows where coffee slowly turns lukewarm near room temperature over time.

🌡️ **Just like the news anchor's weather map:** Red means hot summer heat; blue means cool winter air!

STEP 5 • DATA DETECTIVE

Wait! Why Didn't They Start at the Same Temp?

🔎 The Mystery at Minute 5

If all cups got poured from the exact same coffee pot, why were their starting temperatures different?

- **Insulated (No Lid):** 146.6°F 🔥
- **Insulated (Lid):** 135.5°F ☕
- **Ceramic Mug:** 128.8°F ❄️

1. Staggered Pouring Times

The scientist poured Cup 1 first, then Cup 2, then Cup 3. The pot cooled down while pouring!

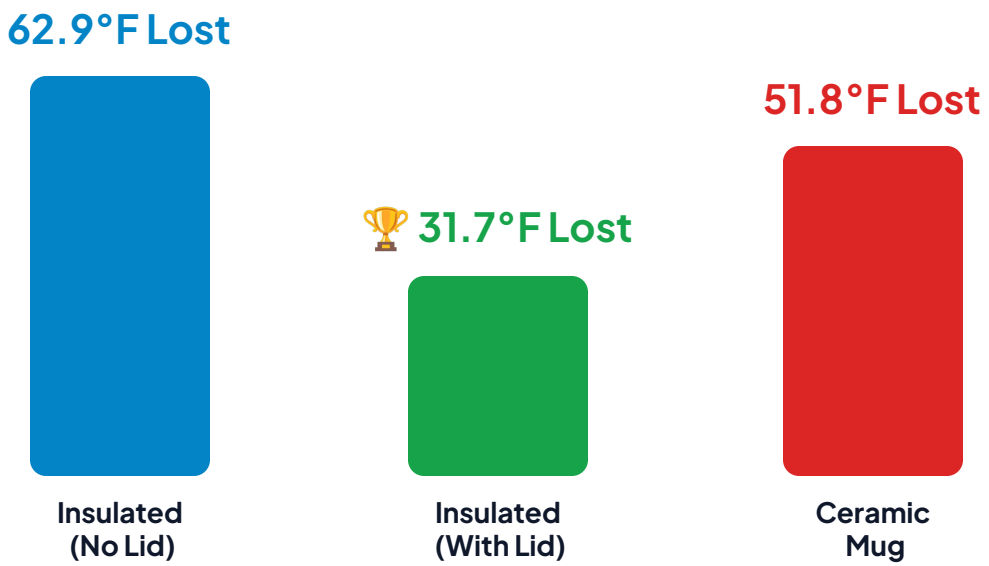
2. Cold Mug Thermal Absorption

The heavy room-temp ceramic mug immediately stole heat from liquid the second it touched the bottom.

🌟 **Lesson:** Great data scientists don't ignore messy data—they explain *why* real life happens!

STEP 6 • FINAL SCOREBOARD

Bar Chart: Total Heat Lost Over 50 Minutes



🎉 The Winner: The lidded cup lost only 31.7°F—keeping coffee nearly 2× warmer than the cup without a lid!

SUMMARY

3 Simple Things We Learned Today

1

Always Put a Lid on It! 🧢

Steam is heat's favorite escape route. Covering the top stops more heat loss than thick side walls alone.

2

Use Different Pictures for Different Questions 📊

Line graphs show direction, scatter plots show raw facts, heatmaps show color patterns, and bar charts show totals!

3

Real World Data is Fun & Messy 🔍

Numbers don't always start out equal, and figuring out why makes you a great data detective.

✨ Now you can explain thermal decay and data visualization to anyone!

🎓 PROJECT COMPLETE!

Pack Your Digital Backpack! 🎒

Want to run this Python code yourself, edit the charts, or check out the dataset? Grab everything inside my public GitHub repository!

Full Python Notebook & Dataset

Includes `hot-coffee.csv`, Matplotlib scripts, and high-res PNG downloads.

Open GitHub Repo

02 / THE BUSINESS PROBLEM

Why Thermal Dynamics Matter

90°C – 95°C

Brewing threshold. Optimal extraction, scalding risk.

55°C – 65°C

The Goldilocks Zone. Peak flavor perception.

< 45°C

Rapid quality decay. Acidic taste profile dominance.

Dataset Structure & Raw Preview (df.head())

```
df.head(5)                                                    Shape: (1200, 5)
```

time_min	temp_c	vessel_type	ambient_temp
0.0	92.5	Ceramic	21.0
1.0	89.8	Ceramic	21.0
2.0	87.2	Ceramic	21.0
3.0	84.9	Ceramic	21.0
4.0	82.7	Ceramic	21.0

Target Variable: temp_c

Continuous float measuring core liquid temperature across 60–minute intervals.

Categorical Factor: vessel_type

Ceramic, Vacuum-Insulated Stainless Steel, and Paper Cup options.

Newton's Law of Cooling

$$T(t) = T_{env} + (T_0 - T_{env}) \cdot e^{-kt}$$

T(t)
Temp at minute t

T_env
Ambient temperature

k
Heat transfer coefficient

05 / PYTHON IMPLEMENTATION

Fitting Non-Linear Decay curve

```python
# SciPy non-linear curve fitting
import numpy as np
from scipy.optimize import curve_fit

def cooling_model(t, k, T_env=21.0, T0=92.5):
    return T_env + (T0 - T_env) * np.exp(-k * t)

# Estimate k parameter for ceramic mug
 popt, _ = curve_fit(cooling_model, df['time'], df['temp'])
 k_est = popt[0]
print(f"Estimated k: {k_est:.4f}")
```

Scipy curve_fit

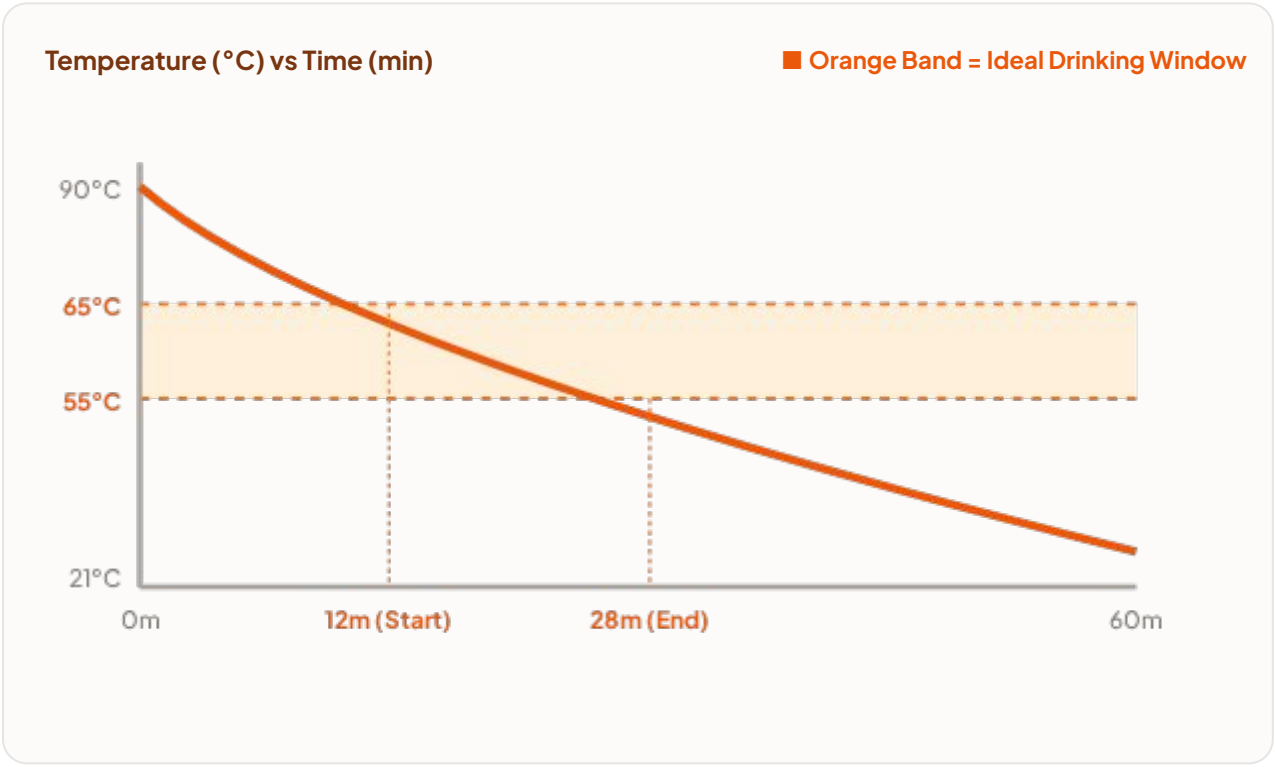
Uses non-linear least squares to fit the decay parameter k.

Key Insight

Ceramic mug exhibits $k \approx 0.042$ min^{-1} under ambient conditions.

06 / VISUAL ANALYTICS

Understanding the Orange Zone (55°C – 65°C)



1. Thermal Safety & Comfort

Above 65°C, liquid causes thermal pain and scalding. Below 65°C, thermal receptors in the tongue comfortably process liquid.

2. Peak Flavor & Volatiles

Aromatic volatile compounds release efficiently without thermal masking, exposing subtle sweetness and acidity.

3. Data Science Goal

Maximize time spent in this 16-minute window (min 12 to 28) using thermal insulation optimization.